Lygtys su keitiniais

Lygis: Baby

Užduotis. Įveskite tinkamą keitinį t, išspręskite gautą pilnąją kvadratinę lygtį ir raskite pradinės lygties realiuosius sprendinius.

1. $\frac{1}{x^2} - 5\frac{1}{x} + 6 = 0.$

2. $x^4 - 5x^2 + 4 = 0.$

3. $(x+1)^4 + 3(x+1)^2 = 4.$

4. $\left(x^2+2x+2\right)^2 - 6\left(x^2+2x+2\right) + 5 = 0.$

5. $x^4 - 4x^2 - 5 = 0.$

6. $\frac{x-2}{x+1} - 2\frac{x+1}{x-2} = 1.$

7. $4^x - 10 \cdot 2^x + 16 = 0.$

8. $\frac{1}{x^2} + \frac{2}{x} - 3 = 0.$

9. $(x-2)^4 - 5(x-2)^2 + 4 = 0.$

10. $2\frac{x-1}{x+2} - \frac{x+2}{x-1} = -1.$

11. $x^6 + 9x^3 + 8 = 0.$

12. $\frac{3x-1}{x+1} + 2\frac{x+1}{3x-1} = 3.$

13. $\left(\frac{x+1}{x-1}\right)^2 - \frac{x+1}{x-1} - 2 = 0.$

14. $4^x + 3 \cdot 2^x - 4 = 0.$

15. $\frac{2}{x^2} + \frac{1}{x} = 1.$

16. $\frac{2x+1}{x-1} - 4\frac{x-1}{2x+1} = 3.$

17. $9^x - 12 \cdot 3^x + 27 = 0.$

18. $x^6 - 7x^3 - 8 = 0.$

19. $\frac{x+1}{x-1} + 6\frac{x-1}{x+1} = 5.$

20. $2(x+3)^4 - (x+3)^2 = 1.$

21. $x^4 + 3x^2 = 4.$

22. $\frac{x+3}{x-2} - 6\frac{x-2}{x+3} = 1.$

23. $\left(x^2+x\right)^2 - 8\left(x^2+x\right) + 12 = 0.$

24. $6\frac{x^2-2x+2}{x^2-2x+3} + 2\frac{x^2-2x+3}{x^2-2x+2} = 7.$

25. $\frac{2}{(x^2+2x+2)^2} + \frac{1}{x^2+2x+2} = 1.$

26. $\left(x^2-4x+2\right)\left(x^2-4x+6\right) = 45.$

27. $6\frac{x^2+2}{x^2+1} + 12\frac{x^2+1}{x^2+2} = 17.$

28. $3 \cdot 9^x - 10 \cdot 3^x + 3 = 0.$

29. $2\frac{2x^2+1}{x^2+2} + 3\frac{x^2+2}{2x^2+1} = 5.$

30. $\left(x^2+3x-4\right)\left(x^2+3x+1\right) = 66.$

31. $\left(\frac{3x+1}{x+1}\right)^2 - 2\frac{3x+1}{x+1} - 8 = 0.$

32. $2x^4 - 7x^2 + 3 = 0.$

33. $2\frac{x^2+3}{x^2+1} + 6\frac{x^2+1}{x^2+3} = 7.$

34. $(2x-1)^4 - 10(2x-1)^2 + 9 = 0.$

35. $2\left(\frac{x-2}{x+1}\right)^2 + \frac{x-2}{x+1} = 1.$

36. $6\frac{x^2-1}{x^2+1} - \frac{x^2+1}{x^2-1} = 1.$

37. $\left(x^2-2x\right)^2 = 3\left(x^2-2x\right) + 4.$

38. $\frac{x^2+2x+2}{x^2+2x+1} + 10\frac{x^2+2x+1}{x^2+2x+2} = 7.$

39. $\frac{3}{(2x^2+1)^2} - \frac{4}{2x^2+1} + 1 = 0.$

40. $\left(x^2-x-2\right)\left(x^2-x+3\right) = 36.$

41. $\frac{x^2+1}{x^2-1} + 6\frac{x^2-1}{x^2+1} = 5.$

42. $10\left(\frac{x}{x^2+1}\right)^2 - \frac{x}{x^2+1} - 2 = 0.$

43. $\left(\frac{1}{4}\right)^x - 5\left(\frac{1}{2}\right)^x + 4 = 0.$

44. $\frac{x^2-3}{x^2-1} + 3\frac{x^2-1}{x^2-3} = -4.$

45. $8x^6 - 9x^3 + 1 = 0.$

46. $\frac{4}{(x^2+1)^2} - \frac{5}{x^2+1} + 1 = 0.$

47. $6\frac{x-1}{x^2-2x+2} - \frac{x^2-2x+2}{x-1} = 1.$

48. $2\left(x^2+x-1\right)^2 + \left(x^2+x-1\right) = 1.$

49. $3\left(\frac{2x-1}{x^2+1}\right)^2 + 2\frac{2x-1}{x^2+1} - 1 = 0.$

50. $10\frac{x}{x^2+1} - 2\frac{x^2+1}{x} = 1.$

Atsakymai

1. $x_1 = \frac{1}{3}, \quad x_2 = \frac{1}{2}.$

2. $x_1 = -2, \quad x_2 = -1.$
$x_3 = 1, \quad x_4 = 2$

3. $x_1 = -2, \quad x_2 = 0.$

4. $x_1 = -3, \quad x_2 = -1.$
$x_3 = 1$

5. $x_1 = -\sqrt{5}, \quad x_2 = \sqrt{5}.$

6. $x_1 = -4, \quad x_2 = \frac{1}{2}.$

7. $x_1 = 1, \quad x_2 = 3.$

8. $x_1 = -\frac{1}{3}, \quad x_2 = 1.$

9. $x_1 = 0, \quad x_2 = 1.$
$x_3 = 3, \quad x_4 = 4$

10. $x_1 = -\frac{1}{2}, \quad x_2 = 4.$

11. $x_1 = -2, \quad x_2 = -1.$

12. $x_1 = 1, \quad x_2 = 3.$

13. $x_1 = 0, \quad x_2 = 3.$

14. $x_1 = 0.$

15. $x_1 = -1, \quad x_2 = 2.$

16. $x_1 = 0, \quad x_2 = \frac{5}{2}.$

17. $x_1 = 1, \quad x_2 = 2.$

18. $x_1 = -1, \quad x_2 = 2.$

19. $x_1 = 2, \quad x_2 = 3.$

20. $x_1 = -4, \quad x_2 = -2.$

21. $x_1 = -1, \quad x_2 = 1.$

22. $x_1 = \frac{1}{3}, \quad x_2 = \frac{9}{2}.$

23. $x_1 = -3, \quad x_2 = -2.$
$x_3 = 1, \quad x_4 = 2$

24. $x_1 = 0, \quad x_2 = 1.$
$x_3 = 2$

25. $x_1 = -2, \quad x_2 = 0.$

26. $x_1 = 2 - \sqrt{7}, \quad x_2 = 2 + \sqrt{7}.$

27. $x_1 = -\sqrt{2}, \quad x_2 = -1.$
$x_3 = 1, \quad x_4 = \sqrt{2}$

28. $x_1 = -1, \quad x_2 = 1.$

29. $x_1 = -2, \quad x_2 = -1.$
$x_3 = 1, \quad x_4 = 2$

30. $x_1 = -5, \quad x_2 = 2.$

31. $x_1 = -3, \quad x_2 = -\frac{3}{5}.$

32. $x_1 = -\sqrt{3}, \quad x_2 = \frac{-\sqrt{2}}{2}.$
$x_3 = \frac{\sqrt{2}}{2}, \quad x_4 = \sqrt{3}$

33. $x_1 = -\sqrt{3}, \quad x_2 = -1.$
$x_3 = 1, \quad x_4 = \sqrt{3}$

34. $x_1 = -1, \quad x_2 = 0.$
$x_3 = 1, \quad x_4 = 2$

35. $x_1 = \frac{1}{2}, \quad x_2 = 5.$

36. $x_1 = -\sqrt{3}, \quad x_2 = \frac{-\sqrt{2}}{2}.$
$x_3 = \frac{\sqrt{2}}{2}, \quad x_4 = \sqrt{3}$

37. $x_1 = 1 - \sqrt{5}, \quad x_2 = 1.$
$x_3 = 1 + \sqrt{5}$

38. $x_1 = -2, \quad x_2 = -\frac{3}{2}.$
$x_3 = -\frac{1}{2}, \quad x_4 = 0$

39. $x_1 = -1, \quad x_2 = 0.$
$x_3 = 1$

40. $x_1 = -2, \quad x_2 = 3.$

41. $x_1 = -\sqrt{3}, \quad x_2 = -\sqrt{2}.$
$x_3 = \sqrt{2}, \quad x_4 = \sqrt{3}$

42. $x_1 = -2, \quad x_2 = -\frac{1}{2}.$
$x_3 = 1$

43. $x_1 = -2, \quad x_2 = 0.$

44. $x_1 = -\sqrt{2}, \quad x_2 = \frac{-\sqrt{6}}{2}.$
$x_3 = \frac{\sqrt{6}}{2}, \quad x_4 = \sqrt{2}$

45. $x_1 = \frac{1}{2}, \quad x_2 = 1.$

46. $x_1 = -\sqrt{3}, \quad x_2 = 0.$
$x_3 = \sqrt{3}$

47. $x_1 = \frac{-1-\sqrt{5}}{2}, \quad x_2 = \frac{-1+\sqrt{5}}{2}.$
$x_3 = 2$

48. $x_1 = \frac{-1-\sqrt{7}}{2}, \quad x_2 = -1$.
$x_3 = 0, \quad x_4 = \frac{-1+\sqrt{7}}{2}$

49. $x_1 = -2, \quad x_2 = 0$.
$x_3 = 3 - \sqrt{5}, \quad x_4 = 3 + \sqrt{5}$

50. $x_1 = -2, \quad x_2 = -\frac{1}{2}.$
$x_3 = 1$

Keitiniai

1. $t = \dfrac{1}{x}$.

2. $t = x^2$.

3. $t = (x+1)^2$.

4. $t = x^2 + 2x + 2$.

5. $t = x^2$.

6. $t = \dfrac{x-2}{x+1}$.

7. $t = 2^x$.

8. $t = \dfrac{1}{x}$.

9. $t = (x-2)^2$.

10. $t = \dfrac{x-1}{x+2}$.

11. $t = x^3$.

12. $t = \dfrac{3x-1}{x+1}$.

13. $t = \dfrac{x+1}{x-1}$.

14. $t = 2^x$.

15. $t = \dfrac{1}{x}$.

16. $t = \dfrac{2x+1}{x-1}$.

17. $t = 3^x$.

18. $t = x^3$.

19. $t = \dfrac{x+1}{x-1}$.

20. $t = (x+3)^2$.

21. $t = x^2$.

22. $t = \dfrac{x+3}{x-2}$.

23. $t = x^2 + x$.

24. $t = \dfrac{x^2-2x+2}{x^2-2x+3}$.

25. $t = \dfrac{1}{x^2+2x+2}$.

26. $t = x^2 - 4x$.

27. $t = \dfrac{x^2+2}{x^2+1}$.

28. $t = 3^x$.

29. $t = \dfrac{2x^2+1}{x^2+2}$.

30. $t = x^2 + 3x$.

31. $t = \dfrac{3x+1}{x+1}$.

32. $t = x^2$.

33. $t = \dfrac{x^2+3}{x^2+1}$.

34. $t = (2x-1)^2$.

35. $t = \dfrac{x-2}{x+1}$.

36. $t = \dfrac{x^2-1}{x^2+1}$.

37. $t = x^2 - 2x$.

38. $t = \dfrac{x^2+2x+2}{x^2+2x+1}$.

39. $t = \dfrac{1}{2x^2+1}$.

40. $t = x^2 - x$.

41. $t = \dfrac{x^2+1}{x^2-1}$.

42. $t = \dfrac{x}{x^2+1}$.

43. $t = \left(\dfrac{1}{2}\right)^x$.

44. $t = \dfrac{x^2-3}{x^2-1}$.

45. $t = x^3$.

46. $t = \dfrac{1}{x^2+1}$.

47. $t = \dfrac{x-1}{x^2-2x+2}$.

48. $t = x^2 + x - 1$.

49. $t = \dfrac{2x-1}{x^2+1}$.

50. $t = \dfrac{x}{x^2+1}$.